Quantification: 699_HYP_HP_z11c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
---------

| measure | value |
|---|---|
| lacuna count | 30 |
| interior lacuna count | 25 |
| frame-edge lacuna count | 5 |

Interior lacunae only: mean, median and sample SD (n = 25)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1946.6 | 971 | 2669.63 | px^2 |
| major_axis_length_px | lacuna shape | 72.9851 | 54.0186 | 48.0087 | px |
| minor_axis_length_px | lacuna shape | 35.3197 | 30.4091 | 19.8543 | px |
| aspect_ratio | lacuna shape | 2.1706 | 2.0007 | 0.8922 | unitless |
| eccentricity | lacuna shape | 0.8087 | 0.8661 | 0.1461 | unitless |
| solidity | lacuna shape | 0.7102 | 0.7052 | 0.0991 | unitless |
| roots_count | headline | 8.36 | 8 | 5.1062 | count |
| ring_length_r30_px | headline | 295.32 | 275 | 188.705 | px |
| ring_length_r60_px | reported | 515.64 | 488 | 356.486 | px |
| owned_length_px | ownership-dependent | 594.101 | 412.706 | 659.155 | px |
| edge_count | ownership-dependent | 32.16 | 24 | 36.5817 | count |
| mean_edge_length_px | ownership-dependent | 18.258 | 17.8995 | 7.7224 | px |
| perimeter_px | normalised | 272.165 | 182.894 | 206.596 | px |
| ring_area_r30_px2 | normalised | 7568.08 | 6958 | 4485.74 | px^2 |
| ring_area_r60_px2 | normalised | 16029.5 | 14086 | 9239.28 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9959 | 1 | 0.017 | unitless |
| in_frame_fraction_r60 | normalised | 0.9863 | 1 | 0.0474 | unitless |
| ring_density_r30 | normalised | 0.0393982 | 0.0427179 | 0.0159429 | px^-1 |
| ring_density_r60 | normalised | 0.0318751 | 0.0340635 | 0.0138127 | px^-1 |
| roots_per_100px_perimeter | normalised | 3.5012 | 3.4797 | 1.9222 | per 100 px |
| ring_attached_length_r30_px | network v2 | 253.2 | 245 | 188.514 | px |
| ring_attached_length_r60_px | network v2 | 384.16 | 380 | 331.981 | px |
| ring_length_w_r30_px | network v2 | 332.142 | 308.137 | 211.514 | px |
| ring_length_w_r60_px | network v2 | 577.375 | 547.688 | 397.309 | px |
| sholl_crossings_r10 | network v2 | 6.8 | 6 | 4.8477 | count |
| sholl_crossings_r20 | network v2 | 7.28 | 6 | 4.3639 | count |
| sholl_crossings_r30 | network v2 | 7.08 | 7 | 4.4527 | count |
| loop_count | loops, report only | 1.64 | 0 | 3.2645 | count |
| loop_edge_share | loops, report only | 0.1042 | 0 | 0.1342 | share |
| ring_width_mean_r30_px | width | 7.5405 | 7.7118 | 1.319 | px |

Per field
---------

| measure | value | units |
|---|---|---|
| analysed_area_px2 | 961876 | px^2 |
| total_skeleton_length_px | 21874 | px |
| canalicular_length_density_per_px | 0.022741 | px^-1 |
| junction_count | 481 | count |
| junction_density_per_px2 | 0.00050006 | px^-2 |
| lacuna_count | 30 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 3.119e-05 | px^-2 |
| skeleton_component_count | 333 | count |
| mean_component_length_px | 65.6877 | px |
| median_component_length_px | 15 | px |
| field_length_density_w_per_px | 0.0253829 | px^-1 |
| field_density_without_flagged_per_px | 0.022051 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 603 | px |
| unattached_component_count | 31 | count |
| canalicular_length_density_attached_per_px | 0.022741 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0233679 | px^-1 |
| loop_count | 54 | count |
| loop_edge_share | 0.1677 | share |
| edge_count | 1282 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.5609 | px |
| width_median_px | 8 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10.7703 | px |
| width_pixels_used | 17109 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 483 | 34.5143 | 18.5791 | 1.8577 | 0.8428 | 0.9028 | -1.3729 | 9.6004 | 721.977 | True |
| 2 | 889 | 60.8664 | 21.5104 | 2.8296 | 0.9355 | 0.7566 | -1.1858 | 41.9595 | 618.117 | False |
| 3 | 971 | 48.2689 | 39.9508 | 1.2082 | 0.5612 | 0.6637 | -1.1736 | 126.652 | 613.53 | False |
| 4 | 1293 | 67.1123 | 29.3143 | 2.2894 | 0.8996 | 0.7085 | -0.9693 | 147.124 | 738.418 | False |
| 5 | 933 | 52.1712 | 30.6068 | 1.7046 | 0.8098 | 0.7272 | -0.9318 | 146.222 | 325.69 | False |
| 6 | 439 | 27.0007 | 22.5757 | 1.196 | 0.5486 | 0.8591 | -1.0184 | 203.804 | 191.597 | False |
| 7 | 437 | 44.3733 | 15.1685 | 2.9254 | 0.9398 | 0.7083 | 1.2486 | 210.419 | 385.556 | False |
| 8 | 8385 | 126.932 | 101.718 | 1.2479 | 0.5982 | 0.7052 | -1.4611 | 277.754 | 204.132 | False |
| 9 | 1905 | 74.7758 | 58.1104 | 1.2868 | 0.6293 | 0.5641 | 1.396 | 301.221 | 621.288 | False |
| 10 | 657 | 45.9581 | 30.4091 | 1.5113 | 0.7498 | 0.5516 | 0.7816 | 290.753 | 563.455 | False |
| 11 | 9083 | 140.536 | 109.35 | 1.2852 | 0.6281 | 0.6544 | -0.1917 | 390.607 | 39.1286 | True |
| 12 | 11096 | 170.603 | 105.697 | 1.6141 | 0.785 | 0.7494 | -0.1521 | 533.955 | 42.6183 | True |
| 13 | 1403 | 54.0186 | 34.896 | 1.548 | 0.7633 | 0.888 | -0.1596 | 368.566 | 862.24 | False |
| 14 | 3093 | 92.0714 | 54.1062 | 1.7017 | 0.8091 | 0.6916 | 1.2856 | 410.259 | 393.304 | False |
| 15 | 1944 | 82.0911 | 39.6724 | 2.0692 | 0.8755 | 0.6506 | 1.2002 | 562.526 | 659.101 | False |
| 16 | 455 | 30.8298 | 20.9014 | 1.475 | 0.7351 | 0.8228 | 1.2448 | 565.978 | 720.266 | False |
| 17 | 1195 | 44.9336 | 38.8022 | 1.158 | 0.5043 | 0.7607 | -1.4455 | 562.059 | 755.791 | False |
| 18 | 2018 | 112.423 | 30.231 | 3.7188 | 0.9632 | 0.6731 | 1.4432 | 571.53 | 456.488 | False |
| 19 | 634 | 48.158 | 30.1352 | 1.5981 | 0.78 | 0.5992 | -1.5103 | 631.199 | 551.279 | False |
| 20 | 1329 | 62.4213 | 28.6247 | 2.1807 | 0.8887 | 0.8819 | 1.2484 | 643.479 | 606.14 | False |
| 21 | 590 | 46.6483 | 23.3159 | 2.0007 | 0.8661 | 0.6095 | -1.4011 | 672.297 | 622.215 | False |
| 22 | 812 | 43.7603 | 30.951 | 1.4139 | 0.7069 | 0.7167 | -1.4386 | 655.233 | 193.442 | False |
| 23 | 451 | 43.3593 | 20.67 | 2.0977 | 0.8791 | 0.6499 | -0.8882 | 692.12 | 316.188 | False |
| 24 | 587 | 59.9418 | 19.0149 | 3.1524 | 0.9484 | 0.5959 | -1.5348 | 698.601 | 352.387 | False |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 25 | 12055 | 251.462 | 78.5981 | 3.1993 | 0.9499 | 0.6372 | -0.5972 | 767.904 | 745.585 | False |
| 26 | 451 | 45.8335 | 13.8387 | 3.312 | 0.9533 | 0.8623 | 0.6712 | 773.745 | 871.692 | False |
| 27 | 2798 | 118.207 | 35.7656 | 3.305 | 0.9531 | 0.8127 | -1.3169 | 814.447 | 954.982 | False |
| 28 | 2396 | 80.3638 | 41.3693 | 1.9426 | 0.8573 | 0.8812 | -1.0185 | 852.972 | 994.821 | True |
| 29 | 14977 | 374.03 | 83.4021 | 4.4847 | 0.9748 | 0.5035 | -1.4304 | 910.923 | 184.857 | True |
| 30 | 2941 | 141.011 | 34.1041 | 4.1347 | 0.9703 | 0.6575 | 1.2599 | 895.57 | 423.196 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 98 | 161 | 41.1421 | 1 | 41.1421 | 5.5126 |
| 2 | 4 | 203 | 249 | 171.569 | 11 | 15.5971 | 7.3657 |
| 3 | 10 | 358 | 604 | 725.027 | 44 | 16.4779 | 8.6418 |
| 4 | 12 | 386 | 697 | 821.826 | 35 | 23.4807 | 8.1517 |
| 5 | 8 | 298 | 437 | 412.706 | 26 | 15.8733 | 9.642 |
| 6 | 6 | 142 | 243 | 90.2843 | 9 | 10.0316 | 8.3226 |
| 7 | 7 | 222 | 488 | 457.505 | 23 | 19.8915 | 9.0224 |
| 8 | 18 | 742 | 1280 | 1556.99 | 92 | 16.9238 | 7.7584 |
| 9 | 15 | 385 | 781 | 811.056 | 56 | 14.4831 | 7.1094 |
| 10 | 10 | 275 | 551 | 605.676 | 35 | 17.305 | 8.2415 |
| 11 | 17 | 546 | 744 | 648.772 | 50 | 12.9754 | 8.1185 |
| 12 | 10 | 245 | 370 | 335.978 | 18 | 18.6654 | 8.7693 |
| 13 | 2 | 209 | 470 | 315.622 | 15 | 21.0415 | 6.8106 |
| 14 | 18 | 813 | 1542 | 3029.25 | 169 | 17.9246 | 7.4887 |
| 15 | 11 | 397 | 533 | 652.872 | 35 | 18.6535 | 7.7118 |
| 16 | 0 | 10 | 10 | 0 | 0 | 0 | 10.3389 |
| 17 | 2 | 121 | 225 | 17.8995 | 1 | 17.8995 | 4.7611 |
| 18 | 13 | 424 | 731 | 726.399 | 35 | 20.7543 | 7.4846 |
| 19 | 11 | 234 | 311 | 389.291 | 24 | 16.2205 | 6.3077 |
| 20 | 2 | 174 | 186 | 33.3137 | 3 | 11.1046 | 8.1266 |
| 21 | 3 | 160 | 225 | 140.154 | 3 | 46.7181 | 5.9762 |
| 22 | 10 | 360 | 596 | 789.713 | 37 | 21.3436 | 8.5944 |
| 23 | 7 | 206 | 372 | 405.363 | 20 | 20.2681 | 7.4959 |
| 24 | 6 | 276 | 556 | 727.701 | 36 | 20.2139 | 8.1101 |
| 25 | 12 | 409 | 639 | 312.078 | 13 | 24.006 | 5.9067 |
| 26 | 5 | 116 | 139 | 99.2548 | 6 | 16.5425 | 5.2586 |
| 27 | 3 | 31 | 102 | 25.8995 | 2 | 12.9497 | 6.0131 |
| 28 | 0 | 0 | 0 | 0 | 0 | 0 | |
| 29 | 20 | 603 | 1219 | 1321.97 | 82 | 16.1216 | 8.5339 |
| 30 | 14 | 432 | 924 | 1535.08 | 74 | 20.7444 | 7.8714 |

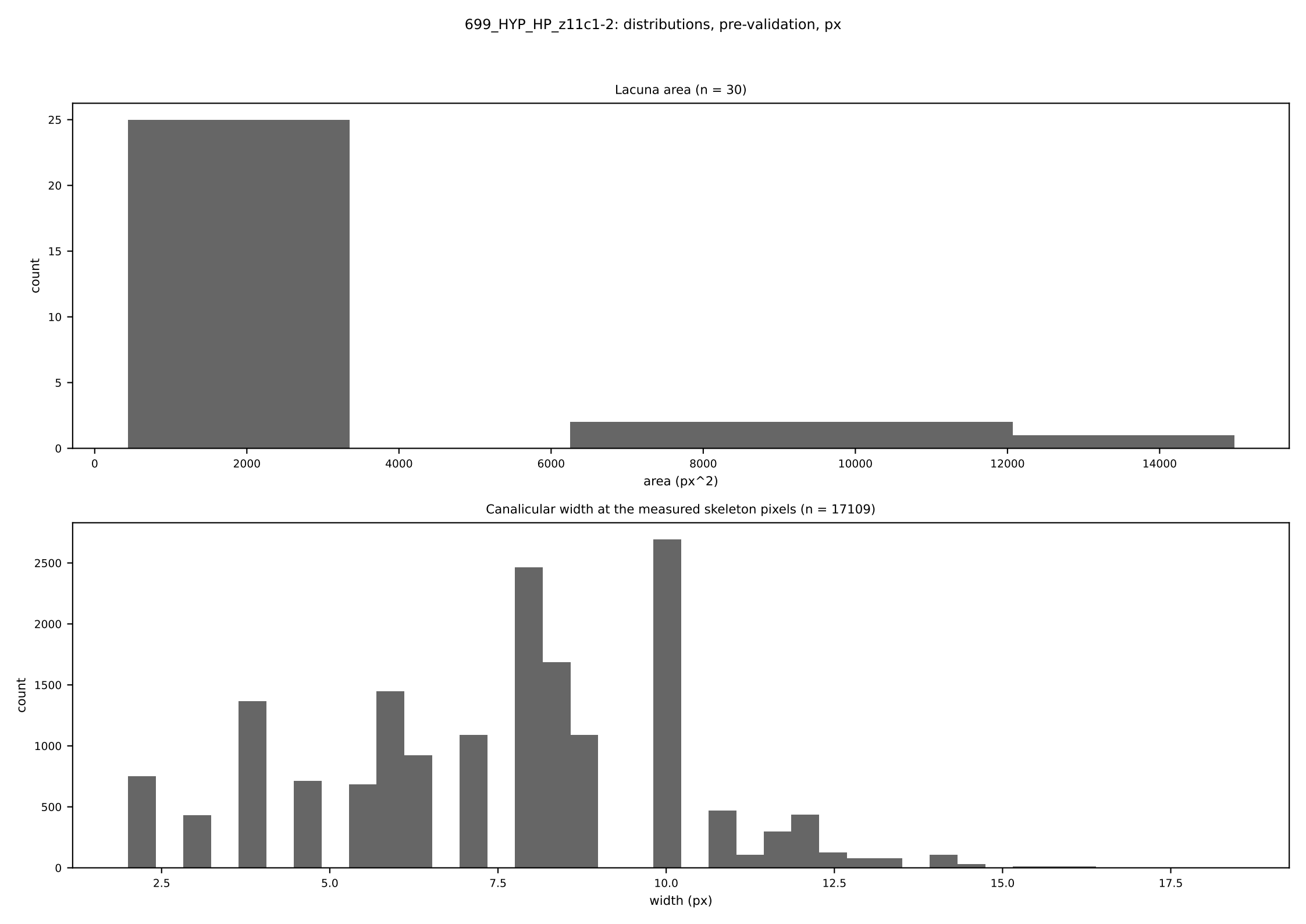
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 95.2548 | 3348 | 7313 | 0.6198 | 0.5747 | 0.0292712 | 0.0220156 | 1.0498 |
| 2 | 172.995 | 6815 | 11086 | 0.9811 | 0.8306 | 0.0297872 | 0.0224608 | 2.3122 |
| 3 | 191.622 | 7249 | 15098 | 1 | 1 | 0.0493861 | 0.0400053 | 5.2186 |
| 4 | 234.25 | 8068 | 19697 | 1 | 1 | 0.0478433 | 0.0353861 | 5.1227 |
| 5 | 172.995 | 6976 | 15405 | 1 | 1 | 0.0427179 | 0.0283674 | 4.6244 |
| 6 | 91.4975 | 3556 | 8201 | 1 | 1 | 0.0399325 | 0.0296305 | 6.5576 |
| 7 | 120.569 | 5863 | 14086 | 1 | 1 | 0.0378646 | 0.0346443 | 5.8058 |
| 8 | 894.833 | 15496 | 28319 | 1 | 1 | 0.0478833 | 0.0451993 | 2.0115 |
| 9 | 394.333 | 8052 | 18251 | 1 | 1 | 0.0478142 | 0.0427922 | 3.8039 |
| 10 | 186.066 | 5049 | 11971 | 1 | 1 | 0.0544662 | 0.0460279 | 5.3744 |
| 11 | 756.861 | 10512 | 16869 | 0.6862 | 0.6501 | 0.0519406 | 0.0441046 | 2.2461 |
| 12 | 695.896 | 10926 | 20640 | 0.7085 | 0.658 | 0.0224236 | 0.0179264 | 1.437 |
| 13 | 174.066 | 7351 | 20270 | 1 | 1 | 0.0284315 | 0.023187 | 1.149 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 14 | 467.446 | 11796 | 28229 | 1 | 1 | 0.0689217 | 0.0546247 | 3.8507 |
| 15 | 318.333 | 8398 | 15415 | 1 | 1 | 0.0472732 | 0.0345767 | 3.4555 |
| 16 | 104.468 | 1694 | 2264 | 1 | 1 | 0.00590319 | 0.00441696 | 0 |
| 17 | 232.285 | 5734 | 14018 | 1 | 1 | 0.0211022 | 0.0160508 | 0.861 |
| 18 | 409.931 | 10384 | 22846 | 1 | 1 | 0.0408321 | 0.0319969 | 3.1713 |
| 19 | 174.853 | 5189 | 9130 | 1 | 1 | 0.0450954 | 0.0340635 | 6.291 |
| 20 | 172.267 | 2782 | 2935 | 1 | 1 | 0.0625449 | 0.0633731 | 1.161 |
| 21 | 155.196 | 4616 | 9759 | 1 | 1 | 0.034662 | 0.0230556 | 1.933 |
| 22 | 182.894 | 6958 | 18413 | 1 | 1 | 0.051739 | 0.0323684 | 5.4676 |
| 23 | 129.539 | 4397 | 10895 | 1 | 1 | 0.0468501 | 0.0341441 | 5.4038 |
| 24 | 172.439 | 5201 | 12627 | 1 | 1 | 0.0530667 | 0.0440326 | 3.4795 |
| 25 | 834.833 | 23174 | 45500 | 1 | 1 | 0.0176491 | 0.014044 | 1.4374 |
| 26 | 109.296 | 4791 | 7177 | 1 | 1 | 0.0242121 | 0.0193674 | 4.5747 |
| 27 | 304.777 | 7140 | 12831 | 0.9164 | 0.8272 | 0.00434174 | 0.0079495 | 0.9843 |
| 28 | 235.48 | 4021 | 7954 | 0.6927 | 0.6449 | 0 | 0 | 0 |
| 29 | 1353.43 | 28693 | 52942 | 0.9016 | 0.8594 | 0.0210156 | 0.0230252 | 1.4777 |
| 30 | 402.333 | 12473 | 26315 | 1 | 1 | 0.0346348 | 0.035113 | 3.4797 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 85 | 99 | 105.184 | 173.64 | 4 | 2 | 2 |
| 2 | 99 | 103 | 225.924 | 274.38 | 5 | 6 | 8 |
| 3 | 329 | 496 | 413.99 | 689.713 | 8 | 8 | 10 |
| 4 | 374 | 621 | 434.049 | 768.115 | 9 | 12 | 12 |
| 5 | 268 | 388 | 345.492 | 503.718 | 6 | 9 | 6 |
| 6 | 96 | 96 | 156.326 | 274.48 | 3 | 3 | 3 |
| 7 | 218 | 380 | 252.551 | 547.688 | 6 | 6 | 6 |
| 8 | 718 | 1062 | 826.541 | 1424.12 | 21 | 13 | 12 |
| 9 | 362 | 604 | 429.948 | 864.269 | 9 | 8 | 11 |
| 10 | 248 | 461 | 308.137 | 608.475 | 5 | 8 | 6 |
| 11 | 513 | 595 | 620.215 | 850.352 | 12 | 14 | 8 |
| 12 | 233 | 277 | 278.108 | 422.061 | 7 | 7 | 2 |
| 13 | 169 | 284 | 233.752 | 529.76 | 4 | 4 | 7 |
| 14 | 799 | 1489 | 916.37 | 1721.76 | 18 | 18 | 17 |
| 15 | 322 | 402 | 445.12 | 600.345 | 10 | 11 | 9 |
| 16 | 10 | 10 | 12.0711 | 12.0711 | 1 | 0 | 0 |
| 17 | 47 | 47 | 132.811 | 244.581 | 3 | 4 | 4 |
| 18 | 400 | 491 | 468.605 | 820.754 | 11 | 13 | 7 |
| 19 | 206 | 223 | 262.894 | 347.007 | 5 | 5 | 4 |
| 20 | 166 | 166 | 199.681 | 215.167 | 4 | 3 | 2 |
| 21 | 118 | 142 | 175.953 | 242.823 | 4 | 4 | 4 |
| 22 | 321 | 490 | 406.563 | 668.286 | 8 | 11 | 9 |
| 23 | 194 | 296 | 238.894 | 424.848 | 4 | 6 | 4 |
| 24 | 254 | 489 | 309.794 | 623.316 | 7 | 5 | 8 |
| 25 | 245 | 294 | 461.274 | 727.767 | 8 | 11 | 8 |
| 26 | 70 | 71 | 135.297 | 159.125 | 1 | 4 | 2 |
| 27 | 26 | 26 | 31.8995 | 112.669 | 0 | 0 | 1 |
| 28 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 29 | 438 | 1131 | 656.103 | 1351.76 | 15 | 20 | 15 |
| 30 | 271 | 473 | 479.605 | 1029.15 | 10 | 10 | 17 |


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 30)
count
area (px^2)
0
2000
4000
6000
8000
10000
12000
14000
0
5
10
15
20
25
Canalicular width at the measured skeleton pixels (n = 17109)
count
width (px)
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
500
1000
1500
2000
2500